

Vaishnavi V Hosagoudar

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EDUCATION

MS Ramiah Institute of Technology

Bachelors of Technology in Electronics and Telecommunication Engineering, GPA 8.39/10

Bengaluru, India

August 2023 - June 2027

GRV Pu College

Puc 2nd year - PCMB , 80%

School Vivekananda

ICSE Board : 89%

TECHNICAL SKILLS

Programming languages: C#, Python, Java, C++

Tools: Visual Studio, Visual Studio Code

Platforms & Frameworks: Node.js, MySQL, MongoDB

PROJECTS

Orga Fuse: Multi-Modal Gen-AI Framework for Organic Produce Authentication

Bengaluru, India

March 2026 - June 2026

- Developed a low-cost, scalable edge AI framework pairing deep learning computer vision with IoT biochemical sensing to authenticate organic produce, successfully replacing destructive and expensive laboratory testing methods.
- Engineered a CycleGAN network to perform unsupervised RGB-to-thermal image translation, synthesizing high-fidelity thermal imagery to uncover hidden subsurface cell-wall irregularities and chemical anomalies.
- Leveraged generative software modeling to detect deep bruising, successfully bypassing the need for cost-prohibitive, specialized thermal imaging hardware.
- Utilized the generative pipeline to mitigate training data scarcity by synthesizing alternative visual modalities, expanding dataset diversity and driving downstream Transformer validation accuracy up to 100%.

Plix: Automate Pill Dispenser

Bengaluru, India

June 2024 - Aug 2024

- Designed and developed an IoT-enabled automated pill dispenser to systematically improve medication adherence and independent living for elderly patients.
- Integrated physical sensors and microcontrollers with a dedicated mobile application to establish real-time dosage tracking, remote scheduling, and continuous patient monitoring.
- Enhanced overall patient safety by designing failsafes that actively mitigate the risks of missed, delayed, or incorrect medical dosages.

Disha: Intelligent Multilingual Route Optimizer using Hybrid TSP Algorithms

Self Project

- Designed and implemented a full-stack logistics route optimization system solving the Traveling Salesman Problem (TSP) using a hybrid algorithmic engine (Nearest Neighbor, Held-Karp, Simulated Annealing).
- Built a Flask backend that integrates Google Maps API for real-time geocoding and distance matrix computation, and dynamically selects optimal algorithms based on delivery node count.
- Developed an interactive React.js frontend with multilingual voice input via the Web Speech API, supporting scalable and user-centric routing for real-world logistics.

Rudra: AI-Powered Personalized Career Guidance System

Self Project

- Developed a full-stack AI-powered career guidance system utilizing a custom-built hybrid AI approach (Python/Flask backend, React frontend), supporting text and voice input with text-to-speech output for enhanced accessibility.
- Leveraged Knowledge Graphs, Rule-Based AI, Multi-Agent Systems, SBERT for semantic matching, and VADER for sentiment analysis to deliver personalized career recommendations and development roadmaps.
- Implemented dynamic web scraping for real-time market trends, creating an explainable and adaptive recommendation engine without reliance on large language models.

ACHIEVEMENTS

Participated in Hackathon

Nov 2024 - Dec 2024

- Developed focused driven solutions, developed a class time table solver, delivering a practical solution and developed problem solving techniques.

Won an IEDA-THON 2025

July 2025 - June 2025

- Designed a sustainable energy solutions, researched on the topic as well as published a paper, demonstrating team-work, problem solving, and innovation, technological solutions.